Intelligent System for Pakistan Sign Language Learning and Translation

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Contents

1. Executive Summary

This business case recommends the development of an Intelligent System for Pakistan Sign Language (PSL) Learning and Translator to address the lack of digital PSL tools in Pakistan and promote inclusion for the Deaf community. The project will deliver a mobile platform with translation, learning, assessment, and dictionary features, enabling affordable and scalable communication support for schools, NGOs, workplaces, and families. The analysis shows that while Year 1 costs are estimated at £25,000–31,000 due to hardware, datasets, and development, annual running costs fall to £6,500–11,600 thereafter, with projected savings of £10,000–14,000 in the first year and £15,000–20,000 annually in later years, achieving breakeven by Year 2–3. Key objectives include enhancing accessibility, reducing interpreter reliance, improving education and employment opportunities, and positioning Pakistan as a pioneer in PSL technology. The findings conclude that the project is both financially viable and socially transformative, with strong alignment to national priorities and the UN Sustainable Development Goals, and should therefore proceed to full development with support from NGOs and government partners.

	Year 1 (£)	**Year 2 (£)**	**Year 3 (£)**	**Year 4 (£)**	**…n (£)**
Total Cash Outlay (Capital + Running)	25,000–31,000	6,500–11,600	6,500–11,600	6,500–11,600	Ongoing running costs
Investment Value – Capital & Revenue	16,700 (capital investment) + 8,000–14,100 (running)	6,500–11,600	6,500–11,600	6,500–11,600	As above
Available / Committed Funding	15,000–16,000	6,000	6,000	6,000	~33,000 secured total (NGO + institutional)
Return on Investment (ROI) (Estimated Savings)	10,000–14,000	15,000–20,000	15,000–20,000	15,000–20,000	Net positive from Year 2–3

2. Introduction and Overview

This Business Case seeks financial approval to deliver an intelligent pakistan sign language learning and translation system that enhances accessibility and independence for the Deaf community. The project addresses the need for cost-effective communication tools by reducing reliance on interpreters while promoting inclusion across education, healthcare, and daily services. Key objectives include developing a working product within 12 months, achieving a significant improvement in communication efficiency, and scaling adoption within a few years after. The investment is expected to generate strong financial and social returns through cost savings and long-term sustainability. The initiative aligns with wider digital inclusion strategies and will be supported through collaboration with NGOs and community stakeholders to ensure cultural accuracy and successful adoption.

3. Market analysis:

Pakistan has a scarce number of digital tools for Pakistan Sign Language (PSL), which leads to exclusion in education, workplaces, and daily interactions for the Deaf community. However, technological readiness has improved substantially in recent years. As advances in artificial intelligence and machine learning, particularly in computer vision and deep learning models, it now makes real-time sign recognition a feasible solution. Despite these advancements, most global research and development efforts have concentrated on American Sign Language (ASL) and British Sign Language (BSL), leaving PSL underrepresented in both academia and technology. This gap presents a unique opportunity. With millions of non-hearing individuals in Pakistan, along with their families, NGOs, and schools seeking better resources, there is a strong demand for innovative solutions that can support communication and learning through PSL

4. Assessment of benefits:

The proposed application offers both tangible and intangible benefits:

Tangibly, it will provide an affordable mobile platform for PSL learning and real-time translation between PSL and spoken or written language. This will reduce reliance on interpreters in educational and professional settings.

Intangibly, the project will have a transformative social impact by promoting inclusion, independence, and accessibility for the Deaf community. It will also facilitate better educational opportunities for Deaf students, cultivate the cultural recognition of PSL, and promote national awareness of the importance of sign language.

5. Cost / Benefits Assessment

The costs of this project remain relatively low as most of the resources needed to develop the project are open source. The primary expenses will include access to cloud storage and hardware required to train and develop deep learning models as well as the time and commitment of the development team. The benefits of the project however, outweigh the costs as in addition to reducing the need for interpreters, it will have a transformative social impact by promoting inclusion, independence, and accessibility for the Deaf community. Furthermore there is potential commercial value from government bodies and NGOs hence future need for software licensing and subscriptions.

1. Financial Cost/Benefit Analysis

Investment and Costs:

Category	Year 1 (£)	Year 2+ (£)	Notes
Capital Investment	16,700	–	Hardware, dataset, licenses, training (one-time)
Running Costs	8,000 – 14,100	6,500 – 11,600	Cloud storage, consultation, maintenance, utilities
Depreciation (hardware/software, 3-year life)	1,700	1,700	Based on £5,000 GPU + £500 software tools
Total Cost (Year 1)	~25,000 – 31,000	–	Includes capital + running

Cash Outlay (Year 1)	~25,000 – 31,000	~6,500 – 11,600	Immediate spend vs ongoing costs

Cost Savings:

Savings Type	First 12 Months (£)	After First Year (£)	Basis
Reduced Interpreter Costs	5,000 – 8,000	7,000 – 10,000	Typical annual interpreter support for institutions
Reduced Training Material Costs (digitized learning vs printed/manual training)	2,000	3,000	Savings on books/manuals/resources
Productivity Gains (faster communication, reduced admin delays)	3,000 – 4,000	5,000+	Valued at staff/student time saved
Total Estimated Savings	**10,000 – 14,000**	**15,000 – 20,000**	Cumulative benefits increase after adoption

2. Unquantified Benefits and Costs

- **Benefits (High Strategic Impact)**
  - **Social Inclusion**: Bridges communication gap for Deaf community.
  - **Education**: Makes PSL accessible to schools/universities at low cost.
  - **Employment**: Enhances workplace inclusion for Deaf employees.
  - **Reputation**: Institution/government adopting project shows commitment to accessibility.
  - **Commercial Potential**: Future licensing to NGOs, schools, and government agencies.

- **Costs (Indirect)**
  - Learning curve for new users (training time).
  - Ongoing model updates (staffing time not always financial).
  - Potential underutilization if adoption is slow.

3. Strategic Contribution

- **Alignment with SDGs**: Quality Education, Reduced Inequalities, Decent Work, and Innovation.
- **Government/NGO Interest**: High potential for partnership and future funding.
- **Scalability**: Can expand beyond PSL to multilingual sign languages (ASL, BSL).
- **Innovation**: Pioneering PSL technology (unique, first-mover advantage).

6. Option appraisal – Recommended option:

Three possible options have been considered. The first option is to take no action, which would mean the exclusion of the Deaf community due to the lack of digital PSL tools. The second option is to rely on manual training programs, which, while useful, have limited reach and scalability. The third and recommended option is the development of an AI-powered mobile application that integrates translation, learning, assessment, and dictionary features. This solution is scalable, cost-efficient, accessible, and capable of creating sustainable impact in education and communication.

7. Key assumptions and dependencies:

The success of the project relies on several key assumptions:

- It assumes the availability of sufficient PSL datasets to support accurate training of the AI models.
- It assumes that users will have access to smartphones and reliable internet connectivity.
- The effectiveness of the system depends on the cultural acceptance of the application by the Deaf community, as accuracy and representation of PSL signs must be validated.
- The project depends on potential collaborations with NGOs, schools, and government institutions for further enhancements and the project's sustainability.

8. Risk and Sensitivity analysis:

There are various risks associated with the project:

Political Risk:
There is a possibility of limited acceptance if the application does not gain the trust of the Deaf community or if stakeholders perceive it as culturally inaccurate. This can be mitigated through early engagement with representative organizations, validation by PSL experts, and active collaboration with NGOs that already work with the community.

Operational Risk:
The effectiveness of the system depends on proper data collection and testing. If PSL datasets are insufficient or inconsistently labeled, recognition accuracy will suffer. Close collaboration with universities and Deaf associations, along with an iterative approach in development, will reduce this risk.

Economic/Financial Risk:
While the project primarily relies on open-source tools, funding will be necessary for hardware resources and ongoing maintenance. Without sufficient external funding or NGO support, scaling may be limited. This risk can be mitigated by pursuing partnerships with NGOs, schools, and grant-making bodies.

Technical Risk:
AI accuracy may initially be low due to the complexity of gesture recognition and limited availability of training data. However, this risk will be managed by using iterative model training, community-driven dataset expansion, and continuous updates to improve accuracy and reliability. Additionally, limited access to smartphones, internet connectivity, and other essential resources for the intended users presents a potential risk.

9. Resource requirements and costs:

Resource requirements and costs	**Estimated Hours**	**Man years (estimated according to 25 days × 8 hrs ×**	**Capital (£)**	**Running costs per annum (range) (£)**

		12 months = 2400 hrs)		
External Consultation	~480 hrs	0.2	2,000	500 – 1,000
Data set (collection + augmentation)	~720 hrs	0.3	3,000	1,000 – 2,000
Hardware (GPU etc.)	~240 hrs	0.1	5,000	1,500 – 2,500
Cloud Storage	~120 hrs	0.05	500	800 – 1,200
Software Management Tools	~120 hrs	0.05	200	100 – 300
Software Licences / Development Tools	~240 hrs	0.1	500	300 – 600
Employee Training	~240 hrs	0.1	1,000	500 – 800
Office Supplies	–	–	200	100 – 200
Office Operational Costs (Transport, Electricity etc.)	–	–	300	500 – 1,000
Research and Development (AI model iterations)	~1,200 hrs	0.5	2,500	1,000 – 1,500
Product Operational Cost (testing, deployment)	~720 hrs	0.3	1,500	1,000 – 2,000
System Updates & Maintenance	~480 hrs	0.2	–	1,200 – 2,000
Total	**~4,560 hrs**	**~1.9 Man-years**	**~16,700**	**~8,000 – 14,100**

10. Funding source / Timing / Certainty:

Initial development of the project will be supported through the academic project framework of FAST NUCES, with team members contributing their time and expertise. For future scaling and sustainability, funding can be sought from NGOs such as Deaf Reach or the Pakistan Association of the Deaf, as well as from government programs focused on accessibility. Additional opportunities may arise through international donors and grants linked to the Sustainable Development Goals (SDGs), particularly those focused on Quality Education and Reduced Inequalities.

11. Timescales:

Summary Milestone Schedule – List key project milestones relative to project start.

Project Milestone	Target Date (mm/dd/yyyy)
• Project Start	09/10/2025
• SRS	10/31/2025
• SDS	12/15/2025
• Authentication and Authorization Module	12/15/2026
• Dictionary Module	12/31/2025
• Translation Module	01/31/2026
• Learning Module	03/10/2026
• Assessment Module	03/10/2026
• Complete Solution Simulation and Testing	04/30/2026
• Deploy Solution	05/30/2026
• Project Complete	06/01/2026

12. Comments / Issues:

One of the main issues that must be carefully managed is cultural accuracy in PSL representation, as errors or misrepresentation could reduce trust in the application. Additionally, ethical considerations in data collection and model training must be respected to ensure that datasets are gathered and used with informed consent.

13. Conclusions and Recommendations

The PSL Translator and Learning App offers a high-impact, socially beneficial, and low-cost solution to a long-standing communication barrier in Pakistan. By leveraging advances in AI and computer vision, the project can deliver a scalable tool that promotes education, accessibility, and inclusion for the Deaf community. It is therefore strongly recommended that this project proceed to full development and that future partnerships and funding sources be pursued to ensure long-term sustainability and adoption.

14. Appendices

Financial Case

The Financial Case demonstrates that the project will result in a fundable and affordable arrangement for the decision-maker. The overall capital and revenue requirements remain relatively low, with Year 1 costs dominated by capital investment in hardware, datasets, and licenses, while subsequent years require only modest running and maintenance costs. The project is affordable and sustainable, with future scalability supported by potential funding from NGOs, government bodies, and academic partners.

Capital and revenue requirements

Description	Value (£)	Start date	End date
Capital Investment (hardware, datasets, licenses, training – one-time)	16,700	Year 1	–
Running Costs (cloud storage, consultation, utilities, maintenance)	8,000 – 14,100 (Year 1) / 6,500 – 11,600 (Year 2+)	Year 1	Ongoing
Depreciation (3-year hardware/software life)	1,700 annually	Year 1	Year 3
Total Cost (Year 1)	~25,000 – 31,000	Year 1	–
Total Cost (Year 2+)	~6,500 – 11,600	Year 2	Ongoing

Resource requirements

Total funding required					
What is it for? (equipment, facilities, external expertise etc)	**When is the cost incurred?**				
	Year 1 2015/16	**Year 2 2016/17**	**Year 3 2017/18**	**Year 4 2018/19**	**Total (£)**
External Consultation	2,000	800	800	800	4,400
Dataset Collection & Augmentation	3,000	1,000	1,000	1,000	6,000
Hardware (GPU, servers)	5,000	–	–	–	5,000
Cloud Storage	500	1,000	1,000	1,000	3,500

Software Tools & Licenses	700	300	300	300	1,600
Employee Training	1,000	500	500	500	2,500
Office & Operational Costs	500	700	700	700	2,600
R&D (AI model iterations)	2,500	1,200	1,200	1,200	6,100
Deployment & Testing	1,500	1,200	1,200	1,200	5,100
System Updates & Maintenance	–	1,500	1,500	1,500	4,500
Total	**16,700**	**8,200**	**8,200**	**8,200**	**41,300**

Funding Currently Secured

Source	**Year 1 (£)**	**Year 2 (£)**	**Year 3 (£)**	**Year 4 (£)**	**Total (£)**
Anticipated NGO/Grant Funding	10,000	4,000	4,000	4,000	22,000
Institutional Support / Academic Partnership	5,000	2,000	2,000	2,000	11,000
Total Secured	**15,000**	**6,000**	**6,000**	**6,000**	**33,000**

Staff Resources

Service Area / Function	**FTEs**	**Year 1 (Q1–Q4)**	**Year 2 (Q1–Q4)**	**Year 3 (Q1–Q4)**
AI Development & R&D	0.5 FTE	Continuous	Continuous	Continuous
Dataset & Annotation	0.3 FTE	Q1–Q3	Q1–Q2	–
Consultation & Community Engagement	0.2 FTE	Q2–Q4	Q1–Q2	As required
Deployment & Testing	0.3 FTE	Q3–Q4	Q2–Q3	Q2–Q3
Maintenance & Updates	0.2 FTE	–	Q3–Q4	Continuous
Total	**~1.5 FTE**	–	–	–

Balance Of Funding Requested

Year	Additional Required (£)
Year 1	10,000 – 16,000
Year 2	2,500 – 5,500
Year 3	2,500 – 5,500
Year 4	2,500 – 5,500
Total	**17,500 – 32,500**

Impact on income and expenditure account

- £10,000–14,000).
- **Year 2–3**: Running costs reduced (~£6,500–11,600 annually) with increased savings (~£15,000–20,000).
- **Year 4–5**: Net positive balance as benefits outweigh ongoing running costs.

Financial benefits

Financial benefits table:

Description	How Calculated	Year 1 (£)	Year 2 (£)	Year 3 (£)	Year 4 (£)	Notes
Reduced Interpreter Costs	Based on typical institutional annual interpreter spend	5,000–8,000	7,000–10,000	7,000–10,000	7,000–10,000	Revenue saving
Reduced Training Material Costs	Digital vs printed/manual training	2,000	3,000	3,000	3,000	Revenue saving
Productivity Gains	Time saved in communication/admin	3,000–4,000	5,000+	5,000+	5,000+	Efficiency gain
Total Savings		**10,000–14,000**	**15,000–20,000**	**15,000–20,000**	**15,000–20,000**	Cumulative benefits

Requirements in order to realise savings

- Sufficient and well-annotated PSL datasets.
- Early adoption by institutions and NGOs.
- User training and awareness campaigns.
- Iterative AI model updates to maintain accuracy.

Non-financial benefits

- **Social Inclusion**: Reduces barriers for the Deaf community.
- **Education**: Affordable PSL resources for schools and universities.
- **Employment**: Workplace inclusivity through better communication.
- **Reputation**: Early adopters seen as accessibility leaders.
- **Innovation & Scalability**: First-mover advantage in PSL AI, with potential to expand to ASL, BSL, and other sign languages.